Learned Soft-Thresholding for Streaming Sparse Signal Recovery in Nuclear Instrumentation

Thiago C. A. Paschoalin, Tiago M. Quirino, João P. B. S. Duarte, and Luciano M. de Andrade Filho

***Abstract*—Experiments in High Energy Physics, such as those conducted at the Large Hadron Collider (LHC), utilize calorimeters for both particle detection and energy measurement. In high-luminosity conditions, these experiments face the challenge of the pile-up effect, where overlapping signals distort energy estimation and hinder trigger system efficiency. Sparse representation (SR)-based methods for calorimeter signal deconvolution and pile-up mitigation are being successfully investigated. This article proposes a novel deconvolutional method that combines a finite impulse response (FIR) filter with SR using a dynamic threshold, estimated by a neural network. Unlike conventional SR approaches that employ iterative methods unsuitable for real-time processing, this solution operates in a non-iterative, feed-forward manner, ensuring compatibility with online data acquisition (DAQ) systems. Performance assessments against traditional least square (LS) techniques and SR methods with fixed thresholds demonstrate enhanced accuracy and resolution. Simulated datasets validate the method's effectiveness, highlighting its capacity to better recover signal characteristics while maintaining the feasibility for hardware implementation.**

***Index Terms*—Calorimetry, data acquisition (DAQ), feedforward neural networks, high energy physics, signal estimation, sparse representation (SR).**

I. Introduction

SIGNAL processing systems in particle physics experiments are essential not only to make data acquisition (DAQ) feasible but also to enable the recovery of the physical properties associated with sensor measurements. Experimental conditions—such as short signal durations, low signal amplitudes in certain events of interest, and background noise—require analog signal shaping to ensure that the resulting waveforms can be sampled at practical rates and remain distinguishable from noise [1], [2]. Furthermore, high event rates often result in signal pile-up, either from background interactions or from valid events occurring in rapid succession. These effects are effectively addressed in the digital processing stage, where dedicated algorithms have proven to be highly advantageous for disentangling pile-up pulses and recovering the true event information [3], [4], [5].

Modern detection systems are designed to sustain high event rates while preserving the ability to identify and select signals online [6]. To achieve this, front-end electronics and digital logic must work in concert to shape, digitize, and process sensor signals under strict latency constraints [7]. Subsystems, such as calorimeters, tracking chambers, and time-of-flight detectors, rely on analog shaping circuits to produce waveforms with predictable structure and duration. These signals, once digitized, are processed by algorithms that extract features like amplitude and timing with high precision, even under conditions of noise and pile-up [4], [8]. Given the large data volumes involved, multi-level trigger architectures discard background and low-priority events early in the acquisition chain [9], [10], ensuring that only the most relevant data are retained. Dedicated hardware platforms, such as field-programmable gate arrays (FPGAs) and application-specified integrated circuits (ASICs), are often employed to meet these online requirements [11], [12]. Several works in the literature have explored algorithmic strategies compatible with these constraints [13].

Duarte et al. [14] addressed online energy estimation for trigger systems using a finite impulse response (FIR) filter based on least square (LS) error minimization, supported by Monte Carlo simulations. The method outperformed traditional approaches, such as matched filters (MFs), autocorrelation filters (AFs), and optimal filters (OFs) [15], [16], [17], offering better accuracy while keeping a simpler implementation. Nonetheless, the limitations of linear methods—particularly under non-Gaussian noise scenarios—have motivated alternative approaches based on sparsity [18].

Sparse representation (SR) methods have shown promise in addressing these limitations by representing signals with minimal non-zero components, effectively reducing artifacts [19]. Barbosa et al. [20] demonstrated improvements using SR for calorimeter readouts but noted that the method is better suited for offline processing, making it unsuitable for real-time trigger systems.

Teixeira et al. [21] proposed an iterative-shrinkage method for energy reconstruction under high pile-up conditions. While effective, its high hardware demands and iterative nature make it less viable for online applications.

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Quirino and de Andrade Filho [22] and Quirino et al. [23] proposed a non-iterative SR-based reconstruction method combining FIR filtering with fixed thresholding, improving accuracy over LS filters while remaining viable for online trigger systems.

However, fixed thresholds limit the full potential of SR methods. Optimal implementations require dynamic thresholding—often computed iteratively [19]—which is incompatible with online constraints. Despite these advances, a gap remains between the reconstruction quality achieved by SR methods and their feasibility in online applications, particularly under high pile-up and signal variability. The method proposed in this work aims to bridge this gap.

This article proposes a novel deconvolutional method that integrates FIR filters with SR and a dynamic shrinkage threshold. A neural network estimates the threshold using calorimeter readout signals and FIR filter outputs, enabling straightforward, streaming processing. The method closely matches the performance of iterative-shrinkage techniques, making it ideal for online applications, such as trigger systems. The technique can also be implemented using FPGA technology.

This strategy preserves the main path of intelligible signal reconstruction through a linear approach based on an FIR filter, while the neural network is employed exclusively to refine the output by enhancing the sparse information that characterizes the data. In addition, this design makes the training process more specialized, since the network learns from structured, physics-driven features rather than from conventional input–output value pairs typically used in standard neural network applications.

The proposed method is evaluated against the LS approach [14] and SR with fixed thresholds [23], demonstrating superior resolution. A neural network designed to estimate the signal amplitude is also considered in the comparison. Simulated datasets with amplitude variation highlight its effectiveness in recovering signal characteristics, minimizing estimation errors, and reducing pile-up effects.

The remainder of this article is organized as follows. Section II presents the calorimeter readout convolutional model, describes sparsity theory, and the proposed energy estimation method. Section III explains the simulated dataset, evaluates performance, and compares results. Section IV concludes this article.

II. SR as a Deconvolutional Processing

When the DAQ system is synchronized to the collision rate, the digitized signal in the calorimeter readout channels ($r[n]$) can be interpreted as the result of a convolution operation ($*$) between a random pulse train of Dirac delta functions [24], representing the energy to be estimated ($x[n]$) from events, and the characteristic pulse of the channel ($h[n]$), derived from the shaping circuit. Equation (1) shows the representation of the signal

$$r[n] = x[n] * h[n] + \sigma[n] \tag{1}$$

where $\sigma[n]$ represents additive white Gaussian noise from the electronic readout chain [14]. Based on this calorimeter model, techniques that perform the inverse process (deconvolution) can be proposed, aiming to estimate the energy $x[n]$ for each event.

Considering a time-windowed output of N samples, the signal is represented by the $\mathbf{r}$ vector, which corresponds to the measured response of the calorimeter over this window. The calorimeter convolutional model can be represented in matrix form, as shown in the following equation:

$$\mathbf{r} = \mathbf{H}\mathbf{x} \tag{2}$$

where $\mathbf{H}$ is a structured matrix formed by the time-shifted versions of the calorimeter's impulse response. It has N rows, corresponding to the number of sampled outputs, and M columns, each representing a shifted version of the reference pulse. Equation (2) is equivalent to (1), but without the additive noise and expressed in a windowed form.

Since $N > M$, the system in (2) is overdetermined. The existence and uniqueness of a solution depend on the rank conditions $\text{rank}(\mathbf{H}) = \text{rank}(\mathbf{H}|\mathbf{r})$ and $\text{rank}(\mathbf{H}) = M$, respectively. Although $\mathbf{H}$ typically has full column rank in our setting, the measurement vector $\mathbf{r}$ contains noise, making it unlikely that $\text{rank}(\mathbf{H}|\mathbf{r}) = \text{rank}(\mathbf{H})$. Thus, the system is generally inconsistent and no exact solution $\mathbf{x}$ satisfying $\mathbf{H}\mathbf{x} = \mathbf{r}$ exists. For this reason, the problem is reformulated to seek an approximate sparse solution whose forward model best fits the observations.

A. Sparsity Theory in Signal Processing

To obtain the sparsest solution, the problem is formulated as an optimization task where the objective is to minimize the ℓ_1-norm of $\mathbf{x}$ as proposed by Elad [19]. The ℓ_1-norm promotes sparsity while ensuring convexity, making the optimization problem computationally efficient. In the presence of noise, a quadratic constraint on the ℓ_2-norm of residue is introduced, ensuring consistency between the estimated and observed signals. The constrained minimization problem is formalized in the following equation:

$$\min_{\mathbf{x}} \|\mathbf{x}\|_1^1 \text{subject to } \|\mathbf{r} - \mathbf{H}\mathbf{x}\|_2^2 \leq \epsilon. \tag{3}$$

The term $\|\mathbf{x}\|_1$ represents the ℓ_1-norm, defined as the sum of the absolute values of the vector $\mathbf{x}$. Similarly, $\|\mathbf{r} - \mathbf{H}\mathbf{x}\|_2^2$ corresponds to the squared Euclidean norm of the residual error, quantifying the difference between the estimated convolved signal and the actual calorimeter measurement. The parameter ϵ defines an upper bound for this error, accounting for measurement noise and model imperfections, particularly due to the presence of white noise, which is not explicitly modeled in the original convolutional system.

A common approach to solving this constrained minimization problem is the introduction of a Lagrange multiplier λ, which reformulates it as an unconstrained optimization problem. This transformation allows the balance between sparsity and fidelity to the measured data, leading to the cost function presented in the following equation:

$$J(\mathbf{x}) = \lambda\|\mathbf{x}\|_1^1 + \frac{1}{2}\|\mathbf{r} - \mathbf{H}\mathbf{x}\|_2^2. \tag{4}$$

If the matrix $\mathbf{H}$ is unitary, then the transformation preserves the Euclidean norm of the signal, and its inverse is given by

$\mathbf{H}^T$. In this case, the residual term $\|\mathbf{r}-\mathbf{H}\mathbf{x}\|_2^2$ can be rewritten in a transformed space, where neighboring samples become decoupled. Defining $\mathbf{x_0} = \mathbf{H}^T\mathbf{r}$, the optimization problem simplifies to the following equation:

$$J(\mathbf{x}) = \lambda\|\mathbf{x}\|_1^1 + \frac{1}{2}\|\mathbf{x_0}-\mathbf{x}\|_2^2. \tag{5}$$

An efficient method to solve this optimization procedure is the use of a shrinkage function $S_\lambda(\cdot)$ [19]. This function is obtained by differentiating the new cost function with respect to $\mathbf{x}$ and setting the derivative to 0, yielding the optimal estimation values. The shrinkage function enforces sparsity by selectively attenuating small values while preserving dominant components. Since particle energy is non-negative ($\mathbf{x} > 0$), the minimization process naturally results in a thresholding effect eliminating values below a given threshold. This property is directly enforced by the ℓ_1-norm penalty. The shrinkage function acts element-wise on the vector $\mathbf{x_0}$, ensuring that values below λ are set to 0 while larger values are shifted downward. The vectorized formulation is given by the following equation, which explicitly captures this effect:

$$\mathbf{x}_\lambda = S_\lambda(\mathbf{x_0}) = \max(\mathbf{x_0}-\lambda, 0). \tag{6}$$

Here, λ is treated as a scalar applied uniformly across all elements of the vector $\mathbf{x_0}$. In component-wise notation, this is expressed as follows, which highlights the individual effect on each sample:

$$x_\lambda[n] = \begin{cases} x_0[n]-\lambda, & x_0[n] > \lambda \\ 0, & x_0[n] \le \lambda. \end{cases} \tag{7}$$

The direct shrinkage function solution is valid only when $\mathbf{H}$ is unitary, which is not the case for the calorimeter model. In practical scenarios where $\mathbf{H}$ is non-unitary, a more general approach is required. To handle the general case, where $\mathbf{H}$ is non-unitary, Elad [19] proposed an iterative soft thresholding algorithm (ISTA) that alternates between estimating $\mathbf{x}$ and applying the shrinkage function. This approach employs a surrogate function, which acts as an approximation to the original cost function, facilitating a more efficient minimization process while ensuring convergence.

Equation (8) illustrates how the estimated amplitude is updated at each iteration. The constant Q is chosen to ensure that the surrogate function is strictly convex. The iterative process continues until the difference between consecutive iterations falls below a predefined threshold ($\mathbf{x}_{k+1}-\mathbf{x}_k < \delta$)

$$\mathbf{x}_{k+1} = S_\lambda\left(\frac{1}{Q}\mathbf{H}^T(\mathbf{r}-\mathbf{H}\mathbf{x}_k)+\mathbf{x}_k\right). \tag{8}$$

The iterative algorithm is well-suited for offline applications, but it poses challenges in real-time environments. However, in high-energy physics experiments with online processing, signal reconstruction must be fast and efficient due to the high collision rates and the large number of channels involved; an iterative approach can become computationally prohibitive. The objective is to develop a non-iterative method that remains grounded in the theory of sparse signals.

B. SR With Dynamic Threshold

To understand the role of threshold λ in the iterative process, we analyze the sequence defined in (8). The first iteration, given by (9), expresses the estimated value as a function of the calorimeter readout $\mathbf{r}$ and its deconvolved counterpart $\mathbf{x_0}$. Expanding this process iteratively reveals that the shrinkage function is applied to a correction term based on the gradient update. This allows us to interpret each iteration as introducing an effective, iteration-dependent threshold $\lambda_k[n]$ that incorporates all iterative corrections. This suggests the existence of an optimal threshold λ_{opt} for each sample, which could be predicted directly, thus eliminating the need for iterative computation

$$\mathbf{x_1} = S_\lambda(\mathbf{\Delta_0}+\mathbf{x_0}) \text{ with } \mathbf{\Delta_0} = \frac{1}{Q}\mathbf{H}^T(\mathbf{r}-\mathbf{H}\mathbf{x_0}). \tag{9}$$

The shrinkage function acts element-wise as shown in the following equation:

$$x_1[n] = \begin{cases} (x_0[n]+\Delta_0[n])-\lambda, & x_0[n]+\Delta_0[n] > \lambda \\ 0, & x_0[n]+\Delta_0[n] \le \lambda. \end{cases} \tag{10}$$

From this expression, we define a modified threshold

$$\lambda_1[n] = \lambda - \Delta_0[n]. \tag{11}$$

It is possible to express the shrinkage function for the first iteration with respect only of $\mathbf{x_0}$, but with the threshold different for every sample

$$x_1[n] = \begin{cases} x_0[n]-\lambda_1[n], & x_0[n] > \lambda_1[n] \\ 0, & x_0[n] \le \lambda_1[n]. \end{cases} \tag{12}$$

The second iteration follows the same structure

$$\mathbf{x_2} = S_\lambda(\mathbf{\Delta_1}+\mathbf{x_1}) \text{ with } \mathbf{\Delta_1} = \frac{1}{Q}\mathbf{H}^T(\mathbf{r}-\mathbf{H}\mathbf{x_1}). \tag{13}$$

Applying again the element-wise representation

$$x_2[n] = \begin{cases} (x_1[n]+\Delta_1[n])-\lambda, & x_1[n]+\Delta_1[n] > \lambda \\ 0, & x_1[n]+\Delta_1[n] \le \lambda. \end{cases} \tag{14}$$

$\mathbf{x_1}$ was defined previously, and can be included in the shrinkage function, which gives us two possibilities depending on the threshold $\lambda_1[n]$: 0 or $x_0[n]-\lambda_1[n]$. A convenient substitution is to replace the null value for $x_0[n]-x_0[n]$.

Equation (15) shows the first possibility where $x_1[n] = x_0[n]-x_0[n]$ and the possible variable changing expressed in (16)

$$x_2'[n] = \begin{cases} (x_0[n]-x_0[n]+\Delta_1[n])-\lambda \\ x_0[n]-x_0[n]+\Delta_1[n] > \lambda \\ 0, \quad x_0[n]-x_0[n]+\Delta_1[n] \le \lambda \end{cases} \tag{15}$$

$$\lambda_2'[n] = x_0[n]-\Delta_1[n]+\lambda \tag{16}$$

$$x_2'[n] = \begin{cases} x_0[n]-\lambda_2'[n], & x_0[n] > \lambda_2'[n] \\ 0, & x_0[n] \le \lambda_2'[n]. \end{cases} \tag{17}$$

If the second possibility of $x_1[n] = x_0[n] - \lambda_1[n]$ is taken, the resulting expression is shown in (18) and the variable change done is expressed in (19)

$$x_2''[n] = \begin{cases} ((x_0[n] - \lambda_1[n]) + \Delta_1[n]) - \lambda \\ x_0[n] - \lambda_1[n] + \Delta_1[n] > \lambda \\ 0, \quad x_0[n] - \lambda_1[n] + \Delta_1[n] \leq \lambda. \end{cases} \tag{18}$$

$$\lambda_2''[n] = \lambda_1[n] - \Delta_1[n] + \lambda \tag{19}$$

$$x_2''[n] = \begin{cases} x_0[n] - \lambda_2''[n], & x_0[n] > \lambda_2''[n] \\ 0, & x_0[n] \leq \lambda_2''[n]. \end{cases} \tag{20}$$

Comparing the two variable changes from (16) and (19), there are differences, but both are non-linear functions of the $\mathbf{x_0}$ and $\mathbf{r}$. Since the goal is not to find that function, it is possible to condense it as a unique $\lambda_2[n]$ that represents that non-linear function output value and is the threshold of the shrinkage function. The final $\mathbf{x_2}$ value is now defined by the step-wise equation

$$x_2[n] = \begin{cases} x_0[n] - \lambda_2[n], & x_0[n] > \lambda_2[n] \\ 0, & x_0[n] \leq \lambda_2[n]. \end{cases} \tag{21}$$

If the iteration process continues until it reaches the stop condition, the estimation of energy values is a shrinkage function that depends on the initial $\mathbf{x_0}$ value and a threshold for each sample, as shown in the following equation:

$$x_k[n] = \begin{cases} x_0[n] - \lambda_k[n], & x_0[n] > \lambda_k[n] \\ 0, & x_0[n] \leq \lambda_k[n]. \end{cases} \tag{22}$$

The threshold vector $\boldsymbol{\lambda}_k$ from a cascade of non-linear operations (shrinkage functions), repeatedly applied with $\mathbf{r}$ and $\mathbf{x_0}$ as arguments. As a consequence, it is not possible to derive a closed-form expression for the sparsest solution to the minimization problem. Therefore, we introduce a function $T : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$, which maps the measurement vector $\mathbf{r}$ and $\mathbf{x_0}$ to a kth iteration threshold vector. We denote the final threshold vector at as a function expressed in the following equation:

$$\boldsymbol{\lambda}_k = T(\mathbf{r}, \mathbf{x_0}). \tag{23}$$

This problem can be approached from a new perspective, where the estimated value $x_k[n]$ is associated with a distinct threshold value $\lambda_k[n]$ for the shrinkage function at each entry. This threshold depends on the neighboring signal from the calorimeter readout channel and the initial energy estimation. A proposed method to determine an effective threshold λ_{ef} for each sample is to use a feed-forward neural network (NN), which takes $\mathbf{r}$ and $\mathbf{x_0}$ as inputs and outputs the threshold for the specific sample being analyzed.

The great advantage of the method is to avoid the iteration process. If the threshold is precisely predicted, the estimated values give the sparsest solution for that application. This can be applied in an online processing scenario.

With this approach introduced, it is important to describe how to obtain the initial energy estimation $\mathbf{x_0}$. This value corresponds to the deconvolved calorimeter readout channel signal and can be calculated using the deconvolutional matrix $\mathbf{C}$, as expressed in (24). The use of the pseudo-inverse matrix

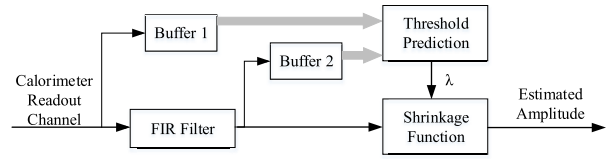

Fig. 1. Proposed method process.

of $\mathbf{H}$ is justified by the fact that the convolutional matrix is not square. Considering the problem in the context of online estimation, the nth sample can be expressed as shown in (25), where $\mathbf{c}_n$ represents the nth row of the $\mathbf{C}$ matrix

$$\mathbf{x_0} = \mathbf{C}r \tag{24}$$

$$\mathbf{C} \approx \mathbf{H}^+ = \left(\mathbf{H}\mathbf{H}^T\right)^{-1}\mathbf{H}$$

$$x_0[n] = \mathbf{c}_n\mathbf{r}. \tag{25}$$

This can be interpreted as an FIR filter, with the order determined by the length of the vector $\mathbf{r}$. High-order filters improve performance in the deconvolutional process but require more computational resources. Since the filter output is processed through a shrinkage function to adjust its value, it is advisable to minimize the filter order while maintaining an accurate solution.

To summarize the overall process, Fig. 1 provides a visual representation. The calorimeter readout channel $\mathbf{r}$ samples are processed through an FIR filter, producing an initial estimation $\mathbf{x_0}$. Both the filtered signal and the original signals are then input into the neural network to predict the effective threshold $\boldsymbol{\lambda}_{ef}$ for the shrinkage function. Finally, the central sample of the processing window is adjusted using the shrinkage function, yielding the final energy estimation through the sparse approach, without requiring an iterative process.

The main effort of the energy estimation still relies on a linear method based on an FIR filter, which is an advantage since it is already established in the literature, considered stable, and with good efficiency when designed correctly. The threshold value generated by the neural network produces a fine-tuning at the estimation values to decrease the remaining energy reconstruction errors, accessing high-order statistics from the non-Gaussian noise inserted by the pile-up effect.

For simplicity, the presented technique will be referred to in this article as the sparse free-running method with dynamic threshold (SFRM-DT).

C. Neural Network Training

An essential step in training NN is the generation of a consistent dataset of effective threshold values, which are used as reference targets ($\lambda_{\text{ref}}[n]$) during supervised learning. These thresholds are derived from the signal behavior and represent practical approximations of the ideal values that ensure correct shrinkage suppression or preservation of the reconstructed amplitudes.

The goal is to determine the effective threshold for input samples so that the shrinkage function in (22) produces the estimated energy ($x_k[n]$). This is achieved by analyzing

observed FIR filter output ($x_0[n]$) and the ground-truth amplitude ($g[n]$), which corresponds to the expected value ($x_\lambda[n]$) after reconstruction.

By inspecting the value of $g[n]$, one can determine which branch of the shrinkage function was applied. When $g[n] \neq 0$, then necessarily $x_0[n] > \lambda[n]$, and the reference threshold can be computed directly from the data as follows:

$$\lambda_{\text{ref}}[n] = x_0[n] - g[n]. \tag{26}$$

This situation corresponds to cases in which the true amplitude is non-zero and preserved by the shrinkage process.

Conversely, if $g[n] = 0$, then the output was suppressed. This may happen under two different conditions: when $x_0[n] \leq 0$ or when $0 < x_0[n] \leq \lambda[n]$. To guide the NN in these cases, the threshold is adjusted relative to the value of $x_0[n]$ using a scaling factor $0 < \alpha < 1$, that provides control over the threshold elasticity and introduces tolerance into the learning targets, which helps the NN generalize. These scenarios require separate treatment because the scaling can stretch or compress the threshold depending on the sign of $x_0[n]$: positive values lead to stretching (increased suppression) and negative values to compression (relaxed suppression).

The two cases of threshold adjustment—compression and stretch—can be described by distinct equations depending on the sign of $x_0[n]$. When $x_0[n] < 0$, the threshold can be compressed toward 0 while maintaining correct suppression, yielding $\lambda_{\text{ref}}[n] = (1-\alpha) \cdot x_0[n]$. Conversely, when $x_0[n] > 0$ but still below the threshold, a slight stretch is recommended to improve robustness, resulting in $\lambda_{\text{ref}}[n] = (1+\alpha) \cdot x_0[n]$. These two expressions can be unified into a single equation by incorporating the absolute value of $x_0[n]$, leading to the more compact form

$$\lambda_{\text{ref}}[n] = x_0[n] + \alpha \cdot |x_0[n]|. \tag{27}$$

This formulation ensures that the threshold is adaptively stretched or compressed in a direction consistent with the sign of the signal estimate, providing both correctness and flexibility in the learning process.

It is possible to combine (26) and (27) to express all cases of the λ_{ref} calculation in a unified piecewise-defined equation, enabling the creation of a consistent reference threshold function across the dataset

$$\lambda_{\text{ref}}[n] = \begin{cases} x_0[n] - g[n], & g[n] \neq 0 \\ x_0[n] + \alpha \cdot |x_0[n]|, & g[n] = 0. \end{cases} \tag{28}$$

With the training dataset defined, NN can be trained to predict the threshold of each sample and apply the shrinkage function that will aim to find the sparsest solution, suitable for an online processing estimation, as employed in trigger systems running on FPGAs.

The training dataset must be composed of simulated signals, as the desired amplitudes are known in this case. For real data estimation, the most effective strategy is to train the neural network using simulated signals and then apply the same parameters to predict λ_{ref}. The performance of the method in this context will depend on how accurately the simulation models the real physical process.

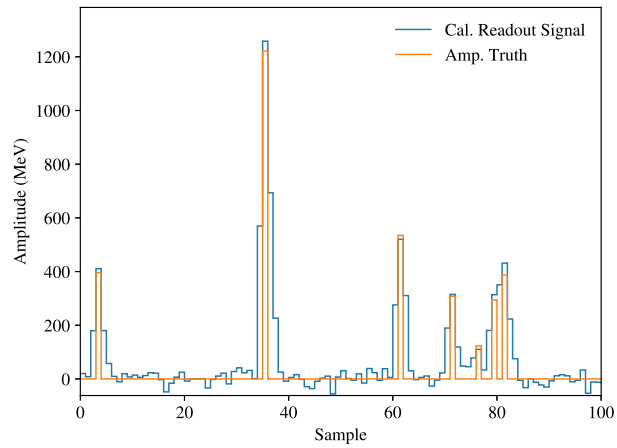

Fig. 2. Extract of data for the calorimeter readout signals and the real amplitude.

III. Results

The results are divided into sections to analyze each component of the method and evaluate the influence of key parameters, including the FIR filter order, the number of neurons in the hidden layer, the input size of the neural network, and the activation functions of the neurons.

The proposed method (SFRM-DT) is compared with the LS FIR filter presented in [14] and the sparse free-running method with fixed threshold (SFRM-FT) described in [23]. A pure neural network that performs amplitude estimation using calorimeter readout samples as input is also included in the analysis. The comparison is based on the root-mean-square error (RMSE) as a measure of estimation accuracy, complemented by plots showing the predicted energies overlaid on the true amplitudes. Both neural network-based approaches employ tenfold cross validation to ensure the evaluation with generalization.

Simulated calorimeter pulses are used to validate the methods, enabling the evaluation of performance in different scenarios, particularly under high event rates.

A. Dataset

Before analyzing the techniques, it is important to describe how the dataset was constructed. The simulated dataset follows the model described by Duarte et al. [14], as it has been adopted in several related studies [20], [21], [22], [23]. An example of the generated data is shown in Fig. 2, where both the calorimeter readout signals and the corresponding true amplitudes to be estimated are illustrated.

All datasets consist of 10 000 samples, with noise modeled as a Gaussian distribution with zero mean and a standard deviation of 20 MeV. The signal amplitude follows an exponential distribution with a mean of 600 MeV. Datasets were generated for different occupancy levels, enabling the evaluation of each technique under conditions ranging from low particle interaction rates to high collision scenarios. A small timing skew is also considered, with a standard deviation of 1 ns and maximum and minimum variations of ± 1.5 ns.

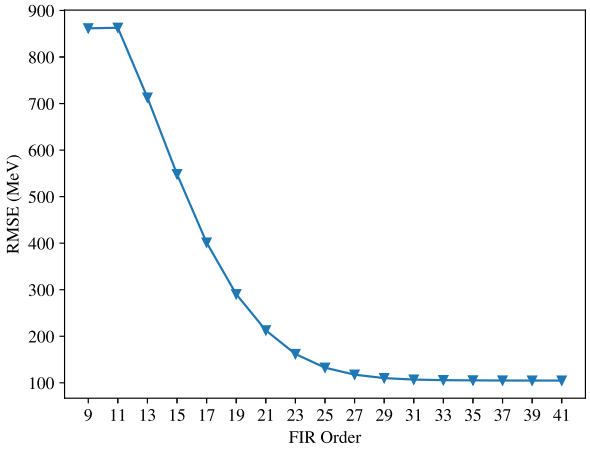

Fig. 3. Relationship between FIR order and RMSE.

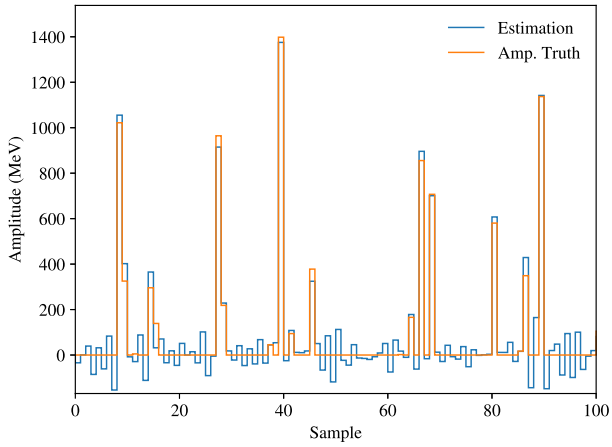

Fig. 4. First estimation samples for the FIR filter for occupancy 20%.

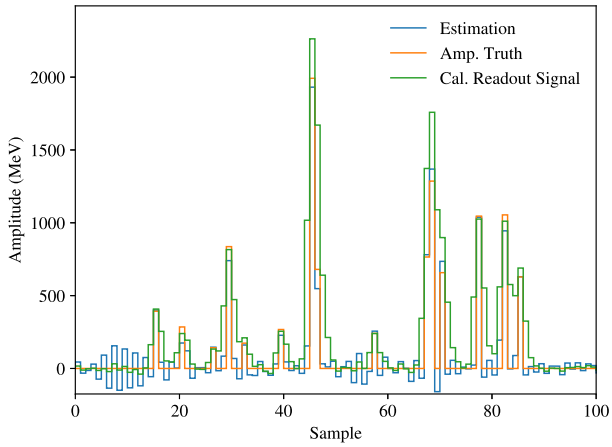

Fig. 5. Samples of the calorimeter readout signal, ground truth, and reconstructed energy.

B. Deconvolutional FIR Filter

The first analysis focuses on the performance of the deconvolutional FIR filter. The filter weights are derived from the rows of the pseudo-inverse of the system convolution matrix, which reaches its maximum value at the center column. The matrix order influences the deconvolutional process, as performance improves with an increased number of samples. However, this improvement is limited, as higher order matrices result in zero elements at the extremities of the FIR weights.

Since the SFRM-DT is designed for real-time processing, minimizing resource usage is critical. Another essential requirement is achieving better performance under high event rates, which forms the basis of the first analysis. A cell occupancy of 80% is considered, and the RMSE value is measured for different FIR filter orders. Fig. 3 illustrates the relationship between the FIR order and the RMSE.

The plot supports the previous statement, showing that the RMSE stabilizes at an FIR order of 37. However, from a hardware perspective, an FIR order of 27 also provides acceptable results. Since the SFRM-DT adjusts the FIR filter output in subsequent steps, the optimal choice for the FIR filter at this stage may not align with the best global solution. A comprehensive analysis of all steps is necessary to identify an effective solution that balances performance with hardware efficiency.

To demonstrate how the proposed method improves results, a dataset with 20% occupancy is analyzed and shown in Fig. 4. This occupancy level was chosen to simplify the visualization of the deconvolutional process under low event rates. The plot highlights undesirable artifacts between real-amplitude collisions, which the shrinkage function aims to suppress.

Fig. 5 illustrates the relationship among the raw calorimeter readout signal, the true amplitude, and the reconstructed energy obtained through the deconvolutional process.

C. Threshold Prediction

The neural network for the threshold prediction includes several parameters that must be adjusted, each of which influences the results. The threshold prediction, when analyzed independently, does not fully represent the best estimate for the scenario, as it only provides the threshold for the shrinkage function. Therefore, several studies are presented to illustrate how different parameter choices and scenarios affect the results.

For the threshold prediction analysis, the occupancy level was set to 20% to ensure the resulting plots are clear and easy to interpret. However, the conclusions can be extended to scenarios with higher event rates.

One factor mentioned that influences the training dataset for threshold prediction is the parameter α. Tests indicate that values between 5% and 20% yield good performance. In this article, this parameter was set to 10%.

To facilitate the hardware implementation, the neural network must be kept as simple as possible. The number of hidden layers was fixed at one to limit the number of weights and, consequently, reduce complexity. Activation functions play a crucial role in simplifying the architecture, as some are significantly easier to implement in hardware and demand fewer resources.

For the hidden layer, the rectified linear unit (ReLU) activation function was chosen due to its simplicity and favorable convergence behavior. A linear activation function was adopted for the output layer, as it supports nega-

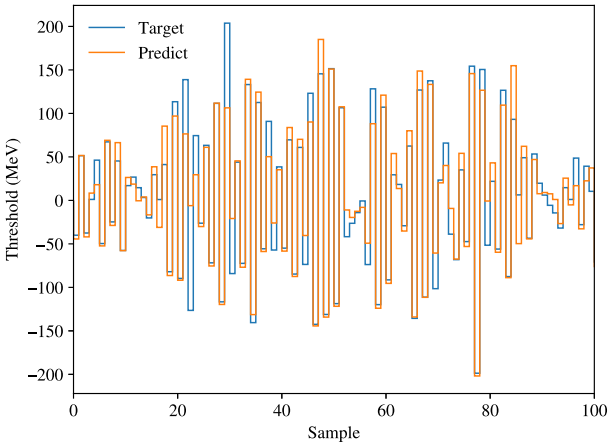

Fig. 6. Threshold prediction.

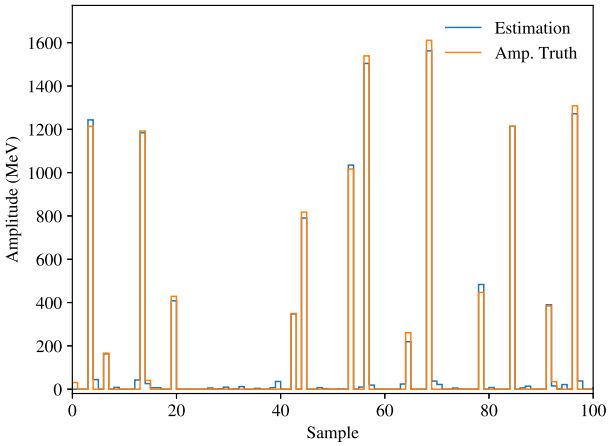

Fig. 7. SFRM-DT estimation for occupancy 20%.

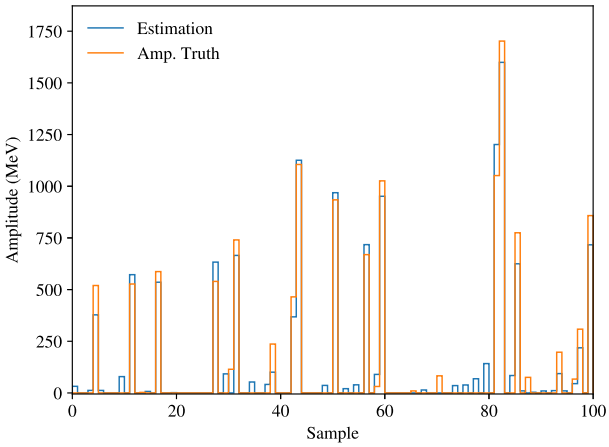

Fig. 8. SFRM-FT estimation for occupancy 20%.

tive values—required for threshold estimation—and is also straightforward to implement.

The number of neurons in the hidden layer and the size of the input vector were optimized to determine the minimum configuration capable of delivering satisfactory results. With five calorimeter readout samples and five FIR outputs as inputs, and ten neurons in the hidden layer, the model achieved an RMSE of 17.13 MeV, as shown in Fig. 6. While this RMSE value alone does not definitively indicate better or worse prediction performance, its practical impact must be assessed within the broader context of the shrinkage function. The key takeaway is the simplicity of the neural network topology achieved, which balances the performance and hardware efficiency.

D. Estimation Results

The next step is to analyze the complete estimation process, which includes the FIR filter, threshold prediction, and shrinkage function. The final amplitude estimation will be compared to the LS FIR, SFRM-FT methods, and a pure neural network that estimates the amplitude.

As mentioned earlier, different parameter configurations affect the results. Therefore, the next estimation analysis will use the best topology, considering both performance and resource efficiency. The FIR filter was configured with 31 weights. For the neural network, the input layer consisted of five calorimeter readout samples and five FIR outputs (totaling ten input parameters). The hidden layer used the ReLU activation function, while the output layer employed a Linear activation function. The hidden layer had ten neurons.

To reach the best performance, the amplitude estimator artificial neural network (ANN) receives 13 calorimeter readout samples as input, followed by a hidden layer with 12 neurons and an ReLU activation function. The output layer uses the same activation function as in the threshold prediction task—linear.

Initially, the occupancy level of 20% was analyzed. The estimation process using the SFRM-DT achieved a mean RMSE of 32.06 MeV. Fig. 7 shows the comparison of the first 100 samples of the real amplitude with the estimated values. The artifacts observed between collision samples in Fig. 4 were eliminated, reducing the RMSE of the deconvolutional process.

For the same occupancy level, the LS FIR method yields an RMSE of 76.09 MeV. Analyzing the estimated samples, it is evident that the higher RMSE, compared to the SFRM-DT, is due to the presence of more nonzero elements between events of interest.

The SFRM-FT achieves an RMSE of 52.86 MeV, as shown in Fig. 8, which illustrates the estimation performance for the first samples. However, this method also fails to filter out unwanted artifacts as effectively as the proposed solution, resulting in a greater number of false amplitudes and consequently a higher RMSE. Overall, the SFRM-DT achieves a 41% reduction in RMSE compared to the best previous solution.

The output samples of the ANN yield a mean RMSE of 34.31 MeV, which is very similar to the result obtained by the SFRM-DT. However, the main drawback of the neural network is its higher resource usage.

To evaluate the scenario with a high event rate, a dataset with 80% occupancy was used. Under these conditions, the SFRM-DT achieved a mean RMSE of 86.48 MeV. The increased error was anticipated due to greater interference

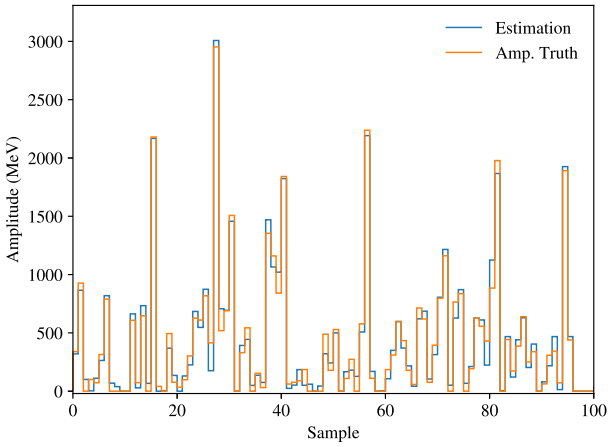

Fig. 9. SFRM-DT estimation for occupancy 80%.

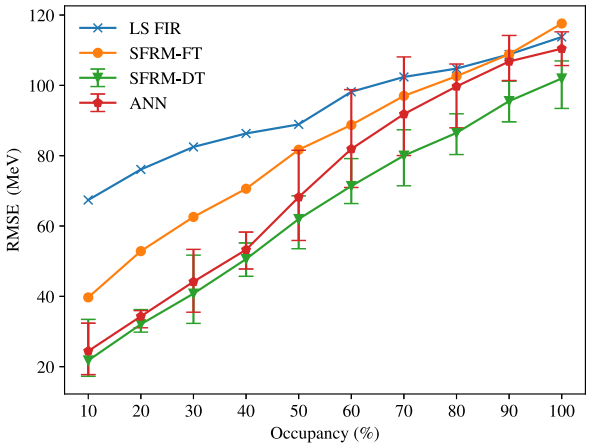

Fig. 10. Performance comparison between the methods.

from neighboring collisions and more pronounced signal corruption.

Fig. 9 illustrates that the estimation process becomes more challenging in this scenario, with fewer zero values compared with the lower occupancy case. This reduced sparsity makes the application of sparse methods more difficult.

Analyzing the LS FIR and SFRM-FT methods, their RMSE values were 104.77 and 102.60 MeV, respectively. The proposed method achieved a reduction of 15.7%, demonstrating better performance. While the difference between approaches is smaller than in the low-occupancy scenario, SFRM-DT still outperforms the alternatives. The neural network achieves a mean RMSE of 99.70 MeV, which is a worse result compared with the 20% occupancy level.

To provide a glimpse of the impact of noise variation, the standard deviation was set to 30 MeV, which is 50% higher than in the previous test. In this new scenario, the SFRM-FT achieved an RMSE of 105.67, compared with 125.22 for the LS FIR, 123.90 for the SFRM-FD, and 110.62 for the neural network prediction. The relative reduction remains almost the same as in the lower noise level.

Fig. 10 shows the comparison of the RMSE values for the four methods across different occupancy levels. The new approach consistently achieves better results in all scenarios, demonstrating its versatility across various environments. Another noteworthy aspect is the simplicity of the neural network, which features a feed-forward structure, a low number of inputs and neurons, and straightforward activation functions, making it suitable for real-time processing and estimation. The ANN needs to be more complex to achieve results comparable with those of SFRM-DT.

IV. Conclusion

The proposed SFRM-DT method achieved better results compared with both the LS FIR and SFRM-FT methods. At low occupancy levels, SFRM-DT reduced the RMSE by 41% compared with SFRM-FT. Although the RMSE reduction at high occupancy levels was smaller due to reduced sparsity, the method still demonstrated superior performance, achieving a 16% reduction in error. The proposed neural network for threshold prediction features a simple feed-forward structure with a small number of neurons and straightforward activation functions, such as ReLU and linear functions. This simplicity, combined with the FIR filters, facilitates the implementation of the method in hardware, such as an FPGA, for use in online trigger systems. Obtaining comparable results demands a more robust amplitude estimator neural network, which in turn increases resource consumption. In addition, the proposed structure uses NN only for fine adjustments, leading the main path reconstruction circuitry to be more predictable.

A. Future Work

Future research will focus on exploring fully machine-learning-based deconvolutional methods to benchmark them against the proposed hybrid FIR+NN approach in terms of both accuracy and resource efficiency. Efforts will also include incorporating more realistic noise models beyond simple Gaussian assumptions, taking into account noise variations and sensitivity to amplitude distributions. In addition, studying the model's impact under further pulse-shaping deformations, such as variations in the time constant and mild undershoot, will be important. Implementation on FPGA and ASIC platforms will be pursued, accompanied by detailed estimates of resource utilization (look-up tables (LUTs) and DSPs) and power consumption. Finally, the methods will be validated with real detector signals, such as LHC test-beam or Run 3 data, to evaluate performance under varying experimental conditions and to address potential domain shifts.

Data and Code Availability

The Python script used in this study is publicly available at: https://github.com/thiagopaschoalin/SFRM-DT (commit `16c0539`).

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